Optimized Traffic Flow at a Single Urban Crossroads : Dynamical Symmetry Breaking

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We propose a stochastic model for the intersection of two urban streets. The traffic state at the crossroads is controlled by a set of traffic lights which periodically switch to red and green with a total period of T. Vehicular dynamics is simulated and total delay experienced by the traffic is evaluated within a definite time interval. Minimising the delay give rises to the optimum signalisation of traffic lights. It is shown that two different traffic phases are identified in the symmetrically loaded crossroads. In the light traffic phase, the green times should be divided to the streets on an equal basis. In the congested phase, Minimising the delay enforces us to break the symmetry between the roads by allocating the majority of green time to one of the roads. In contrast to common sense, equally dividing the green times to the roads, leads to unfairness to drivers.

I. INTRODUCTION

Modelled as a system of interacting particles driven far from equilibrium, *vehicular traffic* provides the possibility to study various aspects of truly non-equilibrium systems which are of current interest in statistical physics [1–4]. For almost half century, physicists have been challenged to understand the fundamental principles governing the vehicular flow [3–5]. Recently, discrete models such as *cellular automata* have provided a significant theoretical framework for modelling traffic flow. The first cellular automata was introduced by *Biham, Middleton* and *Levine*-known as the so-called BML model- which described a simplified network of urban crossroads [6]. soon after, cellular automata, found their way in highway traffic through the pioneer work of *Nagel* and *Schreckenberg* [7] which became the ancestor of many papers in the literature (for a review see ref [1] and the references therein). The BML model itself was later generalized to take into account several realistic features such as faulty traffic lights [8], independent turning of vehicles [9,10], and green-wave synchronization [11]. The Nagel-Schreckenberg and BML model were recently combined to cast an upgraded version of urban network models [12]. In a very recent paper, the model is now extended to account for different types of global signalisation [13]. Despite these efforts and those carried out by traffic engineers, the question of an optimal signalisation scheme for a realistic urban network has not yet been comprehensively reviewed. In the above approaches, the main concern has been focused on the global strategies of the traffic network and frequently the role of isolated crossroads have been suppressed. We believe that the optimisation of traffic flow at a single crossroads is a substantial ingredient towards a global optimisation. Isolated crossings are fundamental operating units of the sophisticated and correlated urban network and thorough analysis of them would be advantageous toward the ultimate task of the global optimisation of the city network. In this regard, our objective in this paper is to analyse the traffic state of an isolated crossroads in order to find a better insight into the problem. In addition to theoretical viewpoint, an investigation of isolated crossroads could be of practical importance. To a very good approximation, marginal crossroads in cities are unaffected by other crossings and can be regarded as isolated ones. There are two basic types of control for traffic lights at intersections: *fixed-cycle* and *traffic-responsive*. Both of these methods can be implemented via centralized or decentralized strategies. The application of each method strongly depends on traffic condition and the topology of the city network. In this paper we study the impact of randomness in vehicular demand at a single crossroads which is controlled under a decentralized fixed-cycle scheme.

II. FORMULATION OF THE MODEL

An isolated crossroads is formed at the intersection of two streets. These streets, in principle, can each carry two oppositely flows of vehicles. Depending on the designing of the crossroads, different phases of movement can be defined (a phase of traffic is defined as the flow of vehicles that proceed a crossroads without conflict). Here for simplicity we restrict ourselves to the simplest structure of a crossroads: a *one-way to one-way* intersection. With no loss of generality, we take the direction of the flow in the first street, hereafter referred to as the street A, northwards. The other street (hearafter referred to as street B) conducts a one-way eastward flow. Cars arrive at the south and west entrances of the crossroads. The traffic flow is controlled by a set of traffic lights. We assume that the traffic is controlled by a fixed-cycle scheme. In this scheme, the lights periodically go green with a fixed period of T. This period is divided into two parts : in the first part, the traffic light is green for street

A and simultaneously red for street B. This part lasts for T_{green} seconds ($T_{green} < T$). In the second part, the lights change colour and the movement is allowed for the cars of road B. The second part lasts from T_{green} to T. This picture is repeated periodically. Cars enter the the crossroads and a fraction of them experience the red light and consequently have to wait until they are allowed to leave the intersection during the upcoming green period. The basic question is

how to adjust the ratio of $\frac{T_{green}}{T}$ *in order to optimise the throughput flow?.*

There is now an almost well-established agreement on the quantitative definition of optimisation. Borrowing from the traffic engineering literature, we adopt *optimised traffic* as a state in which the total delay of vehicles is minimum. In one of our earlier works [14], we analytically evaluated the total delay in terms of arrival as well as that of the exit rates of vehicles. However, our approached was based on the simple assumption of the time-constancy of the arrival rates. In reality, we know that successive cars arrive with fluctuating time-headways which consequently induces time-varying arrival rates. In this paper we address the question of non-constant rates. In order to evaluate the delay, we have simulated the flow of vehicles at the crossroads. For the sake of simplicity, we have assumed that each street has a single lane. For streets with more than one lane, one simply should multiply the value of delay by the number of lanes. Moreover, we allow those fractions of northward (eastward) cars tending to turn right (left) manage to turn via a by-pass road, therefore the cars in our simulation denote those which wish to go through the crossroads without turning. In the next sub section we state our dynamical rules for vehicle movement.

A. Vehicular dynamics at the crossroads

Our crossroads is modelled by a set of two perpendicular chains. Each chain is divided into cells which are the same size as a typical car length . We take the number of cells to be L for both roads. Each cell can be either occupied by a car or being empty. A traffic light periodically changes from red to green. The period of the light is taken to be T units of time and remains constant. The system evolves under a discrete-time dynamics. At each step of time, the system is characterized by the vehicular configuration at each road and the traffic light state of road A (north-bound vehicles). The state of the system at time $t+1$ is updated from that in time t by applying the following rules.

step 1 : signal determination.

We first specify the signal state. If the remainder of division of $t+1$ by T is less than T_{green}, then the signal state is defined as being green for street A and red for street B. Otherwise we set the signal red for street A and street for road B.

step 2 : movement in the green road.

At this stage, we update the position of cars on the green road. This step is divided into two sub-steps. Denoting the position of the i-th cell at time t by pos[i,t], the following half-step rule updates the position of cars synchronously ($i=1$ is the nearest site to the crossroads).

$$pos[i,t+\frac{1}{2}] = pos[i+1,t] \quad i = 1 \cdots L-1 \tag{1}$$

In the above formula, $pos[j,t]=1$ if the site j is occupied and zero otherwise. According to the above rule, each car moves deterministically one cell forward. This type of parallel movement specifies a uniform displacement of cars, i.e., the velocities of all cars are taken to be equal. Nevertheless, in reality, those cars which observe an empty spatial gap may accelerate to fill the gap and increase their chance of going through the green light. To implement this very fact in our model, we have added a probabilistic sub-step to the previous deterministic substep. A car with a space-headway greater than zero, move another extra cell forward with the probability p_{acc} according to the following rule.

$$if \;\; pos[i,t+\frac{1}{2}] = 1 \;\; and \;\; pos[i-1,t+\frac{1}{2}] = 0 \tag{2}$$

then with the probability p_{acc} the following rule occurs.

$$pos[i,t+1] := 0; \qquad pos[i-1,t+1] := 1 \tag{3}$$

In addition, we have assumed that in the green period, no car waits and therefore does not contribute to the delay.

step 3 : movement of red road, delay evaluation
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Similar to the previous step, the updating is divided into two sub-steps. In the first half, we evaluate the delay of cars waiting on the red period of the signalisation. In the second half, we update the position of the moving cars approaching toward the waiting queue. We should note that once the signal switches to red, the moving cars continue their movement until they come to a complete stop upon reaching to the end of the waiting queue. As soon as a car comes to halt, it contributes to the total delay. In order to evaluate the delay, we measure the queue length (the number of stopped cars) at time step t and denote it by the variable Q. We remind that

$$pos[i,t] = 1 \; for \; i = 1 \cdots Q \; and \; 0 \; for \; j = Q+1 \cdots L \tag{4}$$

Delay at time step $t+1$ is obtained by adding the queue length Q to the delay at time step t.

$$delay(t+1) = delay(t) + Q(t) \tag{5}$$

This ensures that during the next time step, all the stopped cars contribute one step of time to the delay. The next sub-step describes the position update of moving cars. Moving cars can potentially be found in the cells $Q+1, Q+2, \cdots, L$. These cars will move one step forward. The possibility of double hopping has been rejected since we have assumed that once a driver notifies the red signal, (s)he no longer attempts to accelerates.

step 4 : entrance of cars to the crossroads.

So far, we have dealt with those cars within the horizon of the crossroads (site L). Now we let the cars enter into this horizon. To fulfill this task, at the end of the movement rules, we evaluate the position of most remote cars on both streets. We denote them by $last_A$ and $last_B$. By definition , $pos_A[j,t] = 0$ for $j > last_A$. A similar statement applies to street B. From our everyday driving experience, we empirically observe that the time head ways between cars randomly varies which consequently implies a random distance headway between successive cars. In order to simulate the entrance behaviour of cars into the horizon of the crossroads, we choose a random integer number between zero and nine. These numbers, representing the distance-headway of the oncoming successor of the farthest car, should be chosen from a realistic random distribution function. Let us denote these headways by h_A and h_B for street A and B respectively. In our simulation, we have tested Gaussian distribution function with a wide range of average as well as standard deviation. Once the random distance headway is chosen, we create a car at the position $last_A + h_A$ ($last_B + h_B$) of the street A (B) respectively. The car creation procedure is valid provided the following constraints are satisfied:

$$last_A + h_A \leq L \quad \text{and} \quad last_B + h_B \leq L \tag{6}$$

If the position of the created car exceeds the horizon length L, the creation is rejected. The above *ad hoc* rules updates the configuration of the crossroads in the next time step. before turning to our results, some points worth mentioning. First, the movement rules of our model differ quantitatively from the existing one in the *highway traffic literature*. In our model, the natural time step is chosen to be the time required for a typical car to replace its predecessor. In contrast to the car-following model of Nagel-Schreckenberg where cars can change their velocity, we have assumed the cars' velocities is almost constant for all vehicles. This assumption is justified by the empirically observed fact that velocity fluctuation is suppressed within the horizon of the crossroads. The situation may differ drastically in high way traffic where one observes a considerable fluctuation in the velocity of cars. Every day measuring of cars velocity in urban areas- where speed-limit regulations e.g. fifty kilometres per hour should be observed by the drivers - supports our assumption of a uniform velocity. The other point to mention concerns the total delay. According to the above rules, no contribution has been devoted to the green phases and we ideally have assumed upon switching the signal from red to green, the waiting cars instantaneously reach to the asymptotic velocity. In reality, however, it takes a few seconds for the waiting queue to react to the signal change. In our model, we have ignored this off-reaction contribution. In the next section, we will explain our simulation results.

III. SIMULATION RESULTS : DYNAMICAL SYMMETRY BREAKING

A. simulation parameters

Before expressing our main results, let us specify the numeric values of the model parameters. The horizon length of the crossroads have been equally taken to be 100 cells for both roads. Setting the bumper-to-bumper distance (7 metres) between successive cars as the cell length, the real horizon length is estimated to be 700 metres. Beyond this distance, it is assumed that drivers do not control the signals and that driving strategy is not affected by the traffic light state. The probability of double hopping is taken to be $p_{acc} = 0.4$. We let the crossroads evolve for 5000 time steps which is approximately equal to a real two-hour period. Our data has been averaged over 50 run of the programme. We let the green time of street A vary from zero to T. For each value of T_{green}, we evaluate the total delay for both streets as well the number of passed vehicle during 5000 time steps.

B. symmetric state: light flow

Here the traffic conditions are equal for both roads. First we discuss the light traffic state. In this case, we equally load the crossroads with entering cars temporally separated by random time-headways from each other. The following graphs depicts the total delay as a function of T_{green} allocated to road A. For the light traffic state which is characterized by large time headway, we observe a flat time interval inside which the total delay remains almost constant. Near the boundary of this area, there is a discontinuous increment of total delay which can be interpreted as a kind of dynamical phase transition. These results can lead to a feasible application by traffic engineers. According to our simulation results, we can set the T_{green} at $\frac{T}{2}$. The existence of an optimum time interval gives us a desirable freedom to account for the probable effect of entrance rate variations. As can be seen from the graphs, increasing the average time-headway leads to a broader optimum interval.

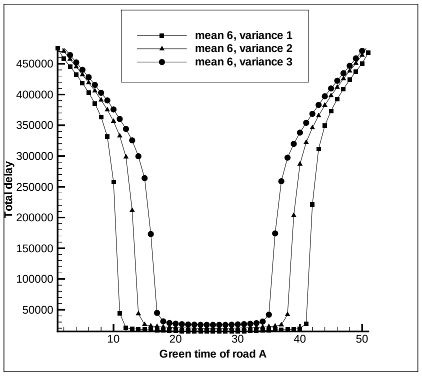

Fig. 1: a symmetric light state. Total delay versus the green time of road A is sketched. As depicted, for fixed average, increasing the variance leads to broadening of the flat region. T is set to 50 and the crossroads has been evolved for 5000 time-step.

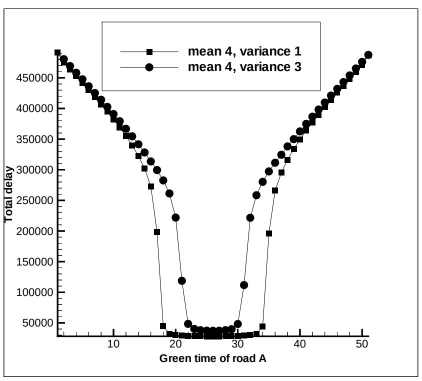

Fig. 2: a more congested state with an average space gap of four cells. The flat region has been shortened in comparison to above graph. The value of T is 50 and the system has evolved for 5000 time-steps.

C. congested flow : symmetry broken phase

Perhaps the most noticeable result of this paper lies in the congested phase of the traffic. For a congested state in which the average time-headway is less than a critical value, the typical sketch of total delay undergoes an abrupt change which is shown in the following set of figures. Unexpectedly, there are two minima for delay which are symmetrically located from each other. The graphs tell us that in order to minimise the total delay we have to choose one of the minima. In other words, we have to trample the right of one direction in favour of the other direction. In the context of statistical physics, this is equivalent to breaking the symmetry of the problem. In practice, we can periodically change the trampled road from A to B. This phenomenon is rather analogous to the spontaneous symmetry breaking in equilibrium statistical mechanics. Creation of two valleys in the delay curve is reminiscent of double-valley creation in the free energy function. In critical phenomena, the physical system will ultimately be found in one of the valleys. In the traffic context, we should signalizes the lights in such a way that-despite the equal conditions for both streets- the majority of green time be allocated to one of streets. Quantitatively, increasing the congestion of the entering cars, leads to a broader height of the barrier. It should be mentioned that the valleys are each discontinuously separated from the barrier. If we decrease the entrance rate, the barrier shrinks to a smaller size and finally at a critical value , it disappears.

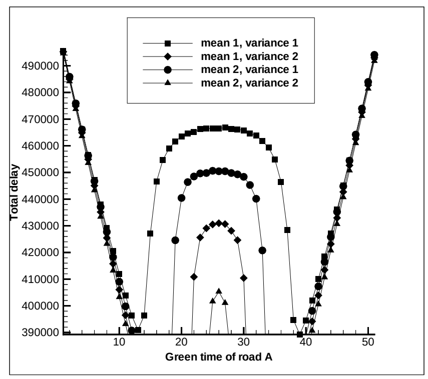

Fig. 3: congested traffic state. The delay curve has two minima and choosing one of them leads to breaking the symmetry between the two roads. The more congested the flow is, the broader the barrier. The system has evolved for 5000 time-steps and T is set to 50.

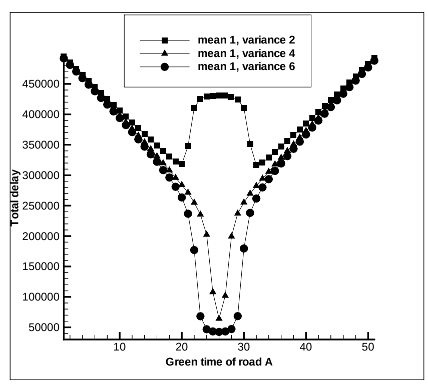

Fig. 4: for fixed value of space gap average, there is a critical variance above which the phase transition between the light and congested state is manifest. For average=1, the critical variance is 4. The system has evolved for 5000 time-steps and T is set to 50.

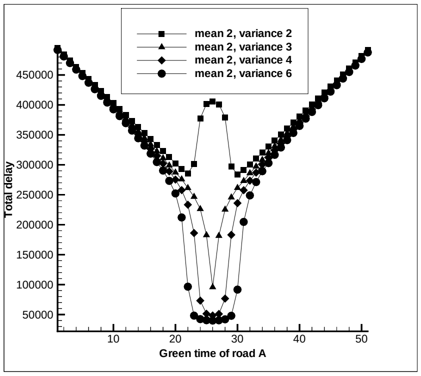

Fig. 5: the effect of increasing the variance for average=2 is shown. The critical variance is 3. the system has evolved for 5000 time-steps and T is 50.

D. Asymmetric case

In the asymmetric states, the entering rates into the roads differ from each other. However, two distinct phases are identified as before. In cases where both roads are carrying light flows, the asymmetry between entering rates causes a displacement on the position of flat optimum region but does not change the overall characteristics of the delay curve. The following figures shows the effect of asymmetry on the delay curve. An interesting asymmetric state is the intersection of a major to a minor street. A large fraction of urban crossroads lie in the category of major-to-minor. The signalisation of these types of crossing is still a controversial subject. The main reason is that in most of intelligent real-time controller systems, the signalisation of these crossroads are highly affected by the major crossroads signalisation schemes which frequently overlook the local optimisation of minor crossroads. In our model, a minor crossing is modelled by a light traffic in one road and a congested one in the other road. The following graphs depicts the behaviour of the delay curve.

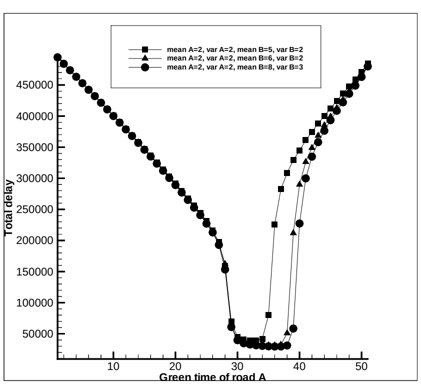

Fig. 6: The asymmetry between roads displaces the flat region toward the road with lighter traffic.

IV. ONE-WAY TO TWO-WAY CROSSROADS

In this section we examine the traffic state at a crossroads where one street is one-way while the other one conducts a two-way flow. We take the traffic flow on the eastward street to be one-way. In one the most popular signalisation schemes, the total cycle period is divided into three movement phases. During the first phase which lasts for T_1 seconds, the traffic on one of roads can move while the lights are red for the other two roads. In the second phase which lasts from T_1 until T_2 the second road's light switches to green and the other two traffic lights are kept red and finally from T_2 until T only the third road is allowed to move. The simulation rules are the same as one-way to one-way rules. The total delay is a function of T, T_1, T_2 as well as inflow rates. The following three dimensional graphs show the behaviour of delay curve as a function of T_1 and T_2. In accordance with the results of previous sections, two distinct states are already identified. In the symmetric state, which corresponds to an overall lightness of the traffic volume, the flat optimal region is now extended to a two dimensional area. The boundary of this area is sharply separated from a highly valued delay area of the delay space. The results warn us to take care in tuning the signalisation times T_1 and T_2 in such a way that we remain inside the flat region. The other state, corresponding to a symmetry broken state, shows the behaviour of the delay curve in congested cases. In the symmetry broken states, three minima valleys are formed and in order to minimise the total delay, one should break the symmetry in favour of one chosen direction. Analogous to one-way crossroads, the valleys are discontinuously separated from each other by a two dimensional barrier.

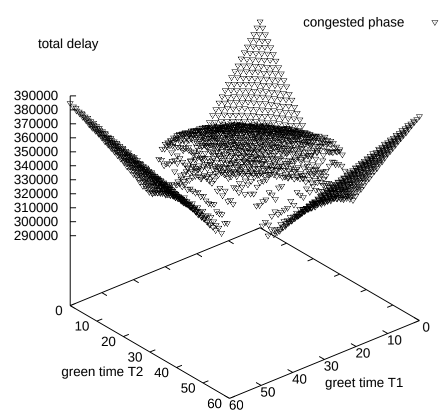

FIG. 1. total delay as a function of green times T_1 and T_2 in a totally symmetric state. the system has evolved for 2000 time-steps and the value of T has been set to 60. the average as well as the standard deviation values of arrival rates distribution functions are two.

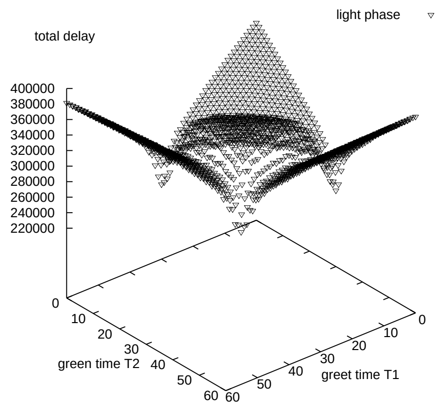

FIG. 2. total delay as a function of green times T_1 and T_2 in a totally symmetric state. the system has evolved for 2000 time-steps and the value of T has been set to 60. the averages are two while the standard deviation of the arrival distribution functions are three. it can be seen that analogous to the one-way to one-way intersection, for average=2, the critical variance is three.

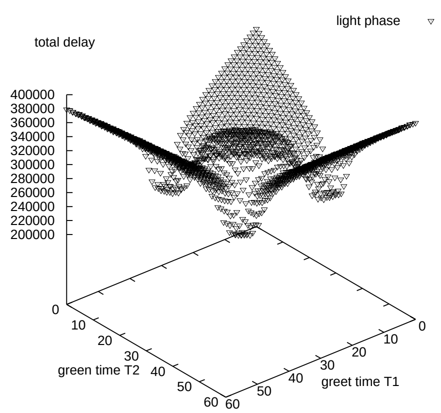

FIG. 3. total delay as a function of green times T_1 and T_2 in a totally symmetric state. the system has evolved for 2000 time-steps and the value of T has been set to 60. the averages are two while the standard deviation of the arrival distribution functions are four.

V. SUMMARY AND CONCLUDING REMARKS

Single crossroads are not just an artificial realization of city network. Marginal urban areas are places where single crossroads are frequently designed and operated. Apart from this convincing fact, it has always been a subject of argument whether to control a crossroads under a centralized or decentralized scheme. In special circumstances, decentralized local adaptive strategies operate more effectively than globally adaptive strategies [16,17,13] and often show a very good performance. Therefore, investigations on single junctions can be of practical relevance for various applications in city traffic. For this purpose, we have developed and analysed a prescription for the traffic light signalisation at a single crossroads. Our investigation predicts the existence of traffic states in which to gain the maximum throughput traffic, one manifestly has to create inequality in favour of one of the flow directions. The phenomenon rather resembles the breaking of the symmetry in the onset of phase transitions in equilibrium statistical physics.
Once the conflicting movement directions are enhanced over two, i.e., one-way to two-way crossing, another question arises which governs the designing the movement structure of the crossroads itself. It can easily be shown that in the case of one-way to two-way crossroads, five different types of signalisation can be defined each of which corresponds to a distinct movement structure. In this paper we have discussed only one of them. Another frequent example is parallel moving of north-bound together with east-bound vehicles toward north in one phase, parallel moving of south-bound together with east-bound vehicles toward south in the second phase and finally the remaining non-conflicting directions in the third phase. Prior to optimisation of green times, one first should determine the optimal movement structure and then optimise the signalisation of the optimal structure. The detailed analysis of the movement structure in one-way to two-way and more generally the intersection of a couple two-way roads will be discussed in a separate paper.

To conclude, we have proposed an optimising adaptive decentralized scheme for a single crossroads on the basis of *minimised total delay* concept. Our next objective is to optimise the traffic flow in a simple urban structure of junctions e.g. a small-sized cluster of crossroads. The results will be reported elsewhere.

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